Modeling with partial differential equations

Exercises – Session 2: Laplace and Poisson equations

Exercise 1: Another proof of the maximum principle in dimension 2

Let $f \in \mathcal{C}^2(\mathbb{R}^2)$ be harmonic on $\mathbb{R}^2$, i.e., $\Delta f = 0$ on $\mathbb{R}^2$. Let $\mathcal{D} = \{(x, y) \in \mathbb{R}^2 \,|\, x^2 + y^2 < r^2\}$ be the open disk of radius $r > 0$ and $\mathcal{C} = \partial\mathcal{D}$. Let $p \in \mathbb{N} \setminus \{0\}$. Let us define the function f_p on $\mathbb{R}^2$ by

$$f_p(x, y) = f(x, y) + \frac{x^2 + y^2}{p}. \tag{1}$$

Question 1 Show that there exists $(a_p, b_p) \in \overline{\mathcal{D}}$ such that

$$f_p(a_p, b_p) = \max_{\overline{\mathcal{D}}} f_p(x, y). \tag{2}$$

Question 2 Show that $(a_p, b_p) \in \mathcal{C}$.

Question 3 Show that there exists $(a, b) \in \mathcal{C}$ such that

$$f(a, b) = \max_{\overline{\mathcal{D}}} f(x, y). \tag{3}$$

Question 4 Show that if f and g are two harmonic functions on $\mathbb{R}^2$ equal on a circle $\mathcal{C}$, then they are equal on the entire disk inside the circle.

Exercise 2: Resolution using Fourier's series

Let $L > 0$ and $f : [0, L] \to \mathbb{R}$ be a continuous function. We consider the following problem:

$$\begin{cases} -u'' = f \text{ on } (0, L), \\ u(0) = a, \\ u(L) = b, \end{cases} \tag{4}$$

where $a, b \in \mathbb{R}$.

We aim to simplify the previous problem by reducing it to a simpler one: the case with homogeneous Dirichlet boundary conditions, i.e.:

$$\begin{cases} -v'' = f \text{ on } (0, L), \\ v(0) = 0, \\ v(L) = 0, \end{cases} \tag{5}$$

As in the lectures, we look for solutions u and v which are in $\mathcal{C}^2([0, L], \mathbb{R})$.

Question 1 Assume there exists a solution v to eq. (5). Define a function $h : [0, L] \to \mathbb{R}$ such that $u = v + h$ is a solution to eq. (4).

Question 2 Show that if $(\lambda, w) \in \mathbb{R} \times \mathcal{C}^2([0, L])$ is a solution to the following problem (with w a non-zero function):

$$\begin{cases} \text{Find } (\lambda, w) \in \mathbb{R}_+^* \times \mathcal{C}^2([0, L]) \text{ such that} \\ -w'' = \lambda w \text{ on } (0, L), \\ w(0) = 0, \\ w(L) = 0, \end{cases} \tag{6}$$

then $(\lambda, \mu w)$ is also a solution for every $\mu \in \mathbb{R}$. Find all the solutions of the previous problem with w such that $\displaystyle\int_0^L w^2 = 1$.

Question 3 Let $N \in \mathbb{N}^*$ and $f(x) = \sqrt{\frac{2}{L}} \sum_{n=1}^{N} f_n \sin\left(\frac{n\pi x}{L}\right)$. Find the solution v to eq. (5).

Exercise 3: Green's Function for the Dirichlet Problem in Dimension 1

Question 1: A fundamental solution of the Laplace operator is defined as a function $\Phi : \mathbb{R} \to \mathbb{R}$ that is in $\mathcal{C}^2(\mathbb{R}^*, \mathbb{R})$ and satisfies, for all $x \in \mathbb{R}^*$,

$$\Phi''(x) = 0,$$

and such that for all $\varphi \in \mathcal{C}_c^2(\mathbb{R})$, the following identity holds:

$$-\int_{-\infty}^{\infty} \Phi(x)\varphi''(x)\,\mathrm{d}x = \varphi(0).$$

Find an expression for a fundamental solution of the form $\Phi(x) = v(|x|)$, where $v : \mathbb{R}_+ \to \mathbb{R}$ is to be determined.

Question 2: Consider the Dirichlet problem on the interval $(0, L)$ with $L > 0$ and $a, b \in \mathbb{R}$:

$$\begin{cases} -u''(x) = f(x) \text{ for } x \in (0, L), \\ u(0) = a, \quad u(L) = b. \end{cases} \tag{7}$$

Derive the expression of the Green's function such that the solution of the Dirichlet problem (7) is given by:

$$u(x) = \int_0^L G(x, y) f(y)\,\mathrm{d}y - \partial_y G(x, L) b + \partial_y G(x, 0) a.$$

Recall that for the Dirichlet problem on the bounded domain $\Omega \subset \mathbb{R}^d$

$$\begin{cases} -\Delta u(x) = f(x) & \text{for } x \in \Omega, \\ u(x) = g(x) & \text{for } x \in \partial\Omega, \end{cases}$$

the Green's function $G : (x, y) \in \Omega^2 \mapsto G(x, y) \in \mathbb{R}$ satisfies:

$$u(x) = \int_\Omega G(x, y) f(y)\,\mathrm{d}y - \int_{\partial\Omega} \partial_n G(x, y) g(y)\,\sigma(\mathrm{d}y).$$

Exercise 4: Green's Function for the Neumann problem in dimension 1

Question 1: Consider the Neumann problem on the interval $(0, L)$ with $L > 0$ and $a', b' \in \mathbb{R}$:

$$\begin{cases} -u''(x) = f(x) \text{ for } x \in (0, L), \\ u'(0) = a', \quad u'(L) = b'. \end{cases} \tag{8}$$

What are the necessary conditions on f, a', b' for the Neumann problem (8) to admit a solution in $\mathcal{C}^2((0, L)) \cap \mathcal{C}^1([0, L])$? Show that if these conditions are satisfied, the solution is uniquely defined up to an additive constant.

Question 2: Define the function

$$N : (x, y) \in (0, L)^2 \mapsto \frac{x^2 + y^2}{2L} - \max\{x, y\} + \frac{L}{3}.$$

Under the conditions established in the previous question, prove that any solution u of (8) satisfies:

$$\forall x \in [0, L], \quad u(x) = \frac{1}{L} \int_0^L u(y) \, \mathrm{d}y + \int_0^L N(x, y) f(y) \, \mathrm{d}y + b' N(x, L) - a' N(x, 0). \tag{9}$$

Exercise 5: Numerical Resolution of the Laplace equation on a square

The goal of this exercise is to compute a numerical solution to the following problem:

$$\begin{cases} -\Delta u = f \text{ in } \Omega, \\ u = g \text{ on } \partial\Omega, \end{cases} \tag{10}$$

where $\Omega = (0, 1) \times (0, 1)$, $f \in \mathcal{C}^2(\Omega)$, and $g \in \mathcal{C}^0(\partial\Omega)$. To achieve this, let $1 \leq I \in \mathbb{N}$ and $h = \frac{1}{I}$. We define a Cartesian grid $(x_{ij})_{0 \leq i,j \leq I}$ by $x_{ij} = (\frac{i}{I}, \frac{j}{I})$. $(I + 1)^2$ is therefore the number of points in the grid, and h is the size of each cell in the mesh.

Part 1

Question 1 Let $h > 0$ and $u \in \mathcal{C}^4([0, 1])$. Show that there exists $C > 0$ such that for every $x \in [h, 1 - h]$,

$$\left| \frac{u(x + h) - 2u(x) + u(x - h)}{h^2} - u''(x) \right| \leq Ch^2. \tag{11}$$

Question 2: Using the previous question, find a way to approximate the value of $-\Delta u$ at x_{ij} for every $1 \leq i, j \leq I - 1$ using the values of u on the grid $(x_{ij})_{0 \leq i,j \leq I}$. Then, provide the full discretized problem using this method.

Part 2

In this section, we consider the case $f(x, y) = 2\pi^2 \sin(\pi x) \sin(\pi y)$ and $g(x, y) = 0$.

Question 3 What is the true solution in this case?

Question 4 To verify that our method is correct, we want to compare our numerical solution to the true solution. Check that the rate of convergence of the method is 2. What is the expression of the error used?

Part 3

In this section, we consider the case $f(x, y) = 0$ and $g(x, y) = xy$.

Question 5 What is the true solution in this case?

Question 6 What is the rate of convergence of our method? Explain.

Part 4

Question 7 What happens if, instead of $-\Delta u = f$, one considers $-\Delta u + c(x, y)u = f$ in the equation, where c is continuous and non-negative? Code and plot the rate of convergence of the method for this case. If there is no analytical solution, the errors can be computed using a reference solution, i.e., a solution computed with a large number of points.